Quantification: 701_z36c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	30
interior lacuna count	18
frame-edge lacuna count	12

Interior lacunae only: mean, median and sample SD (n = 18)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1231.6	804	1554.18	px^2
major_axis_length_px	lacuna shape	57.3435	41.5213	40.4361	px
minor_axis_length_px	lacuna shape	29.4863	24.6873	14.0409	px
aspect_ratio	lacuna shape	1.9179	1.7172	0.7228	unitless
eccentricity	lacuna shape	0.7856	0.8129	0.133	unitless
solidity	lacuna shape	0.7592	0.7626	0.1038	unitless
roots_count	headline	8.1111	7.5	4.1853	count
ring_length_r30_px	headline	346.444	348.5	163.45	px
ring_length_r60_px	reported	794.889	668.5	415.742	px
owned_length_px	ownership-dependent	1549.86	866.047	1655.44	px
edge_count	ownership-dependent	113.444	60	123.81	count
mean_edge_length_px	ownership-dependent	13.5758	13.2988	1.5405	px
perimeter_px	normalised	182.302	137.081	146.446	px
ring_area_r30_px2	normalised	5872.83	4986.5	2988.2	px^2
ring_area_r60_px2	normalised	13210.4	12565.5	6689.24	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9964	1	0.0155	unitless
ring_density_r30	normalised	0.0608118	0.0546013	0.0187195	px^-1
ring_density_r60	normalised	0.0610311	0.0590269	0.0151047	px^-1
roots_per_100px_perimeter	normalised	5.4558	4.563	2.6634	per 100 px
ring_attached_length_r30_px	network v2	282.833	298	150.143	px
ring_attached_length_r60_px	network v2	532.056	429	404.673	px
ring_length_w_r30_px	network v2	391.432	392.164	187.824	px
ring_length_w_r60_px	network v2	897.836	753.192	475.11	px
sholl_crossings_r10	network v2	7.5556	7.5	3.2399	count
sholl_crossings_r20	network v2	8.8333	7.5	3.9295	count
sholl_crossings_r30	network v2	9.8889	9.5	4.7883	count
loop_count	loops, report only	10.7778	3	16.6081	count
loop_edge_share	loops, report only	0.2349	0.1809	0.2141	share
ring_width_mean_r30_px	width	6.0777	6.0628	1.2979	px

Per field
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measure	value	units
analysed_area_px2	956749	px^2
total_skeleton_length_px	57414	px
canalicular_length_density_per_px	0.0600095	px^-1
junction_count	1833	count
junction_density_per_px2	0.00191586	px^-2
lacuna_count	30	count

lacunae_per_px2	3.136e-05	px^-2
skeleton_component_count	831	count
mean_component_length_px	69.0903	px
median_component_length_px	13	px
field_length_density_w_per_px	0.0675598	px^-1
field_density_without_flagged_per_px	0.0660626	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	92	px
unattached_component_count	7	count
canalicular_length_density_attached_per_px	0.0600095	px^-1
field_length_density_with_unattached_per_px	0.0601056	px^-1
loop_count	315	count
loop_edge_share	0.2831	share
edge_count	4493	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.2558	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	38781	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	4432	103.113	57.3862	1.7968	0.8308	0.8752	0.7863	35.4294	265.904	True
2	5447	154.686	50.1635	3.0836	0.946	0.7894	0.5665	62.9728	466.817	True
3	7233	130.507	74.6116	1.7492	0.8205	0.8879	1.2582	34.0104	969.455	True
4	22983	288.728	111.574	2.5878	0.9223	0.7629	1.3196	89.9404	130.262	True
5	13014	273.142	75.2869	3.628	0.9613	0.7172	1.0206	198.948	86.2458	True
6	1625	83.1546	31.6031	2.6312	0.925	0.7835	0.114	172.29	1011.39	True
7	804	41.5213	33.4623	1.2408	0.592	0.6949	-0.5169	176.293	670.124	False
8	437	31.4519	18.3155	1.7172	0.8129	0.8918	1.4887	285.545	929.915	False
9	843	61.4626	24.6873	2.4896	0.9158	0.7274	1.4958	308.81	200.344	False
10	1362	78.3413	37.8814	2.0681	0.8753	0.558	0.4269	332.607	242.698	False
11	406	28.9527	20.3619	1.4219	0.7109	0.7823	-0.2801	325.505	634.365	False
12	1410	54.1218	34.645	1.5622	0.7683	0.921	-1.5551	413.697	828.063	False
13	9556	142.266	91.5717	1.5536	0.7653	0.9091	1.4115	461.649	963.906	True
14	481	29.6358	22.2683	1.3309	0.6599	0.8698	1.5129	580.47	966.104	False
15	487	49.3817	14.2888	3.456	0.9572	0.7945	1.4407	585.704	1001.05	True
16	498	39.5856	18.2761	2.166	0.887	0.7626	0.4797	621.745	804.205	False
17	6510	186.09	58.4873	3.1817	0.9493	0.6768	-1.566	684.663	733.562	False
18	428	48.7828	12.3797	3.9405	0.9673	0.7897	1.2696	681.061	1004.84	True
19	2377	86.3084	62.7585	1.3752	0.6865	0.5865	-0.3235	741.99	862.039	False
20	994	72.7028	20.0045	3.6343	0.9614	0.7412	1.4325	768.803	413.569	False
21	1166	44.1476	35.3039	1.2505	0.6004	0.9276	0.2819	791.344	1008.62	True
22	537	37.0303	20.8186	1.7787	0.827	0.792	1.4611	859.722	497.698	False
23	897	53.6979	24.7285	2.1715	0.8877	0.766	-1.5247	884.094	479.08	False
24	485	29.7344	25.3636	1.1723	0.5219	0.7555	-0.793	918.491	106.885	False

25	433	29.5156	20.2362	1.4586	0.728	0.8625	0.4576	943.721	446.413	False
26	3880	120.945	42.3369	2.8567	0.9367	0.9282	-1.4866	1004.7	384.256	True
27	2460	98.2976	35.4294	2.7745	0.9328	0.9004	1.5163	1009.82	777.268	True
28								153.691	294.607	False
29								200.671	282.601	False
30								593.976	873.082	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	288	585	186.953	16	11.6846	6.5032
2	14	683	1606	2529.28	178	14.2094	6.5137
3	11	429	835	1819.53	145	12.5485	6.7812
4	23	634	734	516.446	42	12.2963	4.6297
5	17	516	702	432.333	36	12.0093	4.4728
6	9	333	676	647.115	49	13.2064	6.1216
7	8	443	1213	1932.6	158	12.2317	5.8682
8	7	291	894	1276.12	86	14.8386	7.4536
9	6	134	201	103.811	8	12.9764	4.8623
10	11	389	841	397.605	30	13.2535	6.9731
11	4	230	655	98.0833	8	12.2604	5.8913
12	6	365	917	816.541	52	15.7027	4.7742
13	9	454	858	624.245	52	12.0047	6.4637
14	4	291	428	224.125	18	12.4514	6.2613
15	4	152	172	104.74	7	14.9629	5.1202
16	7	159	406	74.0416	7	10.5774	4.6982
17	16	670	1500	1905.83	133	14.3295	4.6364
18	6	276	546	551.961	42	13.1419	6.4452
19	14	439	1050	4008.34	274	14.629	4.0253
20	17	714	1616	3304.89	247	13.3801	6.9086
21	6	284	662	846.968	61	13.8847	6.0351
22	2	320	606	314.296	18	17.4609	7.6508
23	10	365	682	3722.98	279	13.344	6.519
24	8	445	1363	5625.82	435	12.9329	7.9631
25	9	349	611	915.553	68	13.464	8.0706
26	6	489	1002	1996.95	158	12.6389	7.2113
27	4	260	778	207.267	12	17.2723	
28	4	143	332	2487.91	168	14.809	
29	5	141	347	163.74	13	12.5954	4.7029
30	8	348	646	525.132	40	13.1283	6.0628

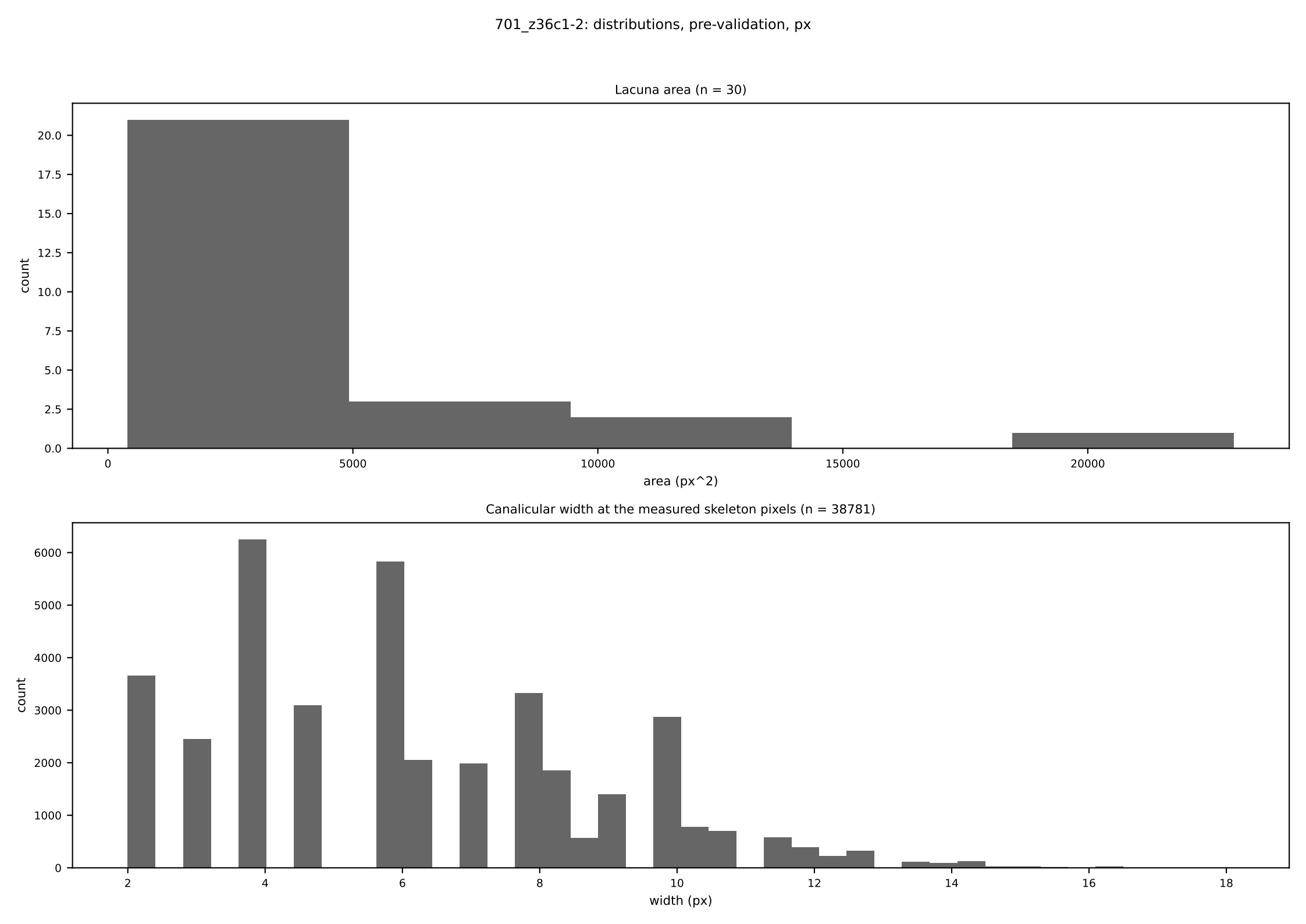
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	375.375	5789	9786	0.7168	0.6687	0.0497495	0.0597793	2.1312
2	544.968	12171	26429	0.8379	0.7759	0.056117	0.0607666	2.569
3	492.416	6624	12045	0.4791	0.4258	0.0647645	0.0693234	2.2339
4	1258.65	16541	17950	0.8467	0.7108	0.038329	0.0408914	1.8274
5	909.744	13435	18845	0.8275	0.7843	0.0384072	0.0372513	1.8687
6	303.842	5339	10334	0.5936	0.5634	0.0623712	0.0654151	2.9621
7	227.635	6849	19116	1	1	0.064681	0.0634547	3.5144
8	94.468	5255	15574	1	1	0.0553758	0.0574034	7.4099
9	173.794	3164	5692	1	1	0.0423514	0.0353127	3.4524
10	267.019	7373	16497	1	1	0.0527601	0.050979	4.1196
11	105.332	5383	16362	1	1	0.0427271	0.0400318	3.7975
12	158.267	6781	16160	1	1	0.0538269	0.056745	3.7911
13	551.263	10755	17854	0.7333	0.694	0.0422129	0.0480565	1.6326

14	101.225	3856	5783	1	0.9344	0.0754668	0.07401	3.9516
15	121.64	2578	3234	0.7213	0.6493	0.0589604	0.0531849	3.2884
16	139.823	4563	8762	1	1	0.0348455	0.0463365	5.0063
17	628.95	14793	28532	1	1	0.0452917	0.0525726	2.5439
18	124.368	4363	8006	0.7149	0.6458	0.0632592	0.0681989	4.8244
19	461.47	10803	21825	1	1	0.0406369	0.04811	3.0338
20	198.309	7886	19993	1	1	0.0905402	0.0808283	8.5725
21	148.125	4251	10059	0.6223	0.5852	0.0668078	0.0658117	4.0506
22	134.338	3706	8642	1	1	0.0863465	0.0701227	1.4888
23	179.622	4408	8457	1	1	0.082804	0.0806432	5.5672
24	118.953	5728	17030	1	1	0.0776885	0.0800352	6.7253
25	97.8823	4099	7159	1	1	0.0851427	0.0853471	9.1947
26	315.907	6243	12280	0.6169	0.5909	0.0783277	0.0815961	1.8993
27	302.007	5690	14075	0.5535	0.5397	0.0456942	0.0552753	1.3245
28	65.8406	2874	5474	1	1	0.0497564	0.0606503	6.0753
29	56.4264	3472	7172	1	1	0.0406106	0.0483826	8.8611
30	72.0833	4718	9557	1	1	0.0737601	0.0675944	11.0983

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	262	472	317.794	668.6	9	7	6
2	597	1129	769.571	1831.75	16	14	18
3	421	700	487.019	945.11	12	8	9
4	461	470	699.742	814.725	22	12	9
5	411	421	579.973	785.796	13	15	13
6	320	493	371.664	758.955	9	7	7
7	372	884	508.617	1375.79	11	9	18
8	274	593	338.22	1003.15	7	9	9
9	64	64	149.61	223.451	2	6	4
10	326	351	449.132	962.063	9	10	7
11	165	265	261.794	735.926	5	7	8
12	214	356	409.534	1031.16	10	7	14
13	388	518	513.304	957.524	11	14	11
14	203	220	323.38	479.262	7	7	11
15	143	154	170.782	193.267	5	4	1
16	75	75	173.297	455.919	2	8	4
17	494	1014	752.754	1686.11	13	19	16
18	232	342	307.38	608.032	6	8	6
19	319	478	483.576	1172.22	11	12	14
20	640	1463	821.955	1855.73	12	17	18
21	234	500	323.25	745.713	6	7	8
22	314	426	353.338	669.333	8	7	10
23	355	594	413.421	770.458	8	6	8
24	425	1345	511.546	1557.21	10	11	12
25	334	576	396.007	696.63	5	9	10
26	475	843	541.019	1118.6	12	14	10
27	176	405	282.966	878.838	7	10	13
28	125	298	158.983	372.35	5	4	3
29	110	143	152.297	386.463	5	5	3
30	282	432	388.321	727.813	6	6	9


701_z36c1-2: distributions, pre-validation, px
Lacuna area (n = 30)
count
area (px^2)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
5000
10000
15000
20000
Canalicular width at the measured skeleton pixels (n = 38781)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16
18